

Hun-Seok Kim

其他

基本信息

姓名: Hun-Seok Kim

所属院校: 密歇根大学-安娜堡

地区: United States

学科领域

通信工程

人工智能

电子与计算机工程

集成电路设计与集成系统

电子信息工程

微电子科学与工程

研究领域

无线通信与智能天线

深度学习与神经网络

低功耗电路与能量管理

物联网通信与M2M

低功耗设计与能耗优化

计算机体系结构

信息论与编码技术

通信系统与调制解调

射频集成电路

嵌入式系统与硬件设计

+1

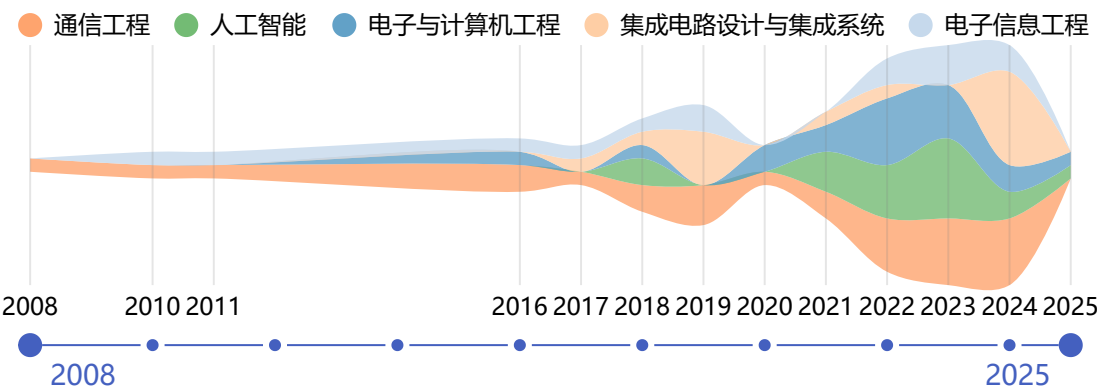
个人简介

Hun-Seok Kim is an esteemed faculty member at the University of Michigan. His research focuses on digital communication, machine learning, VLSI architecture, and signal processing. He has authored numerous publications, including works on programmable deep-learning processors, sparse matrix multiplication accelerators, and innovative approaches to wireless communication and brain-machine interfaces. His contributions to the field are recognized through his impactful research and academic engagements. Email: Verified email at umich.edu.

工作经历

教育经历

近年发表的论文



H-PCC: Point Cloud Compression With Hybrid Mode Selection and Content Adaptive Down-Sampling

IEEE ROBOTICS AND AUTOMATION LETTERS

Liu, Bowen; Chen, Yu; Wang, Boyang; Yang, Mingyu; Kim, Hun-Seok

被引用次数: 0 2025

FALCON: An FPGA Emulation Platform for Domain-Specific SoCs (DSSoCs)

IEEE DESIGN & TEST

Krishnakumar, Anish; Yu, Hanguang; Ajayi, Tutu; Goksoy, A. Alper; Pandey, Vishrut; Mack, Joshua; Hassan, Sahil; Chen, Kuan-Yu; Chakrabarti, Chaitali; Bliss, Daniel W.; Akoglu, Ali; Kim, Hun-Seok; Dreslinski, Ronald G.; Blaauw, David; Ogras, Umit Y.

被引用次数: 0 2024

Canalis: A Throughput-Optimized Framework for Real-Time Stream Processing of Wireless Communication

ACM TRANSACTIONS ON RECONFIGURABLE TECHNOLOGY AND SYSTEMS

Chen, Kuan-yu; Nelson, Thomas mason; Khadem, Alireza; Fayazi, Morteza; Singapuram, Sanjay sri vallabh; Dreslinski, Ronald; Talati, Nishil; Kim, Hun-seok; Blaauw, David

被引用次数: 0 2024

DAP: A 507-GMACs/J 256-Core Domain Adaptive Processor for Wireless Communication and Linear Algebra Kernels in 12-nm FINFET

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Chen, Kuan-Yu; Yang, Chi-Sheng; Sun, Yu-Hsiu; Tseng, Chien-Wei; Fayazi, Morteza; He, Xin; Feng, Siying; Yue, Yufan; Mudge, Trevor; Dreslinski, Ronald; Kim, Hun-Seok; Blaauw, David

被引用次数: 0 2024

AIMMI: Audio and Image Multi-Modal Intelligence via a Low-Power SoC With 2-MByte On-Chip MRAM for IoT Devices

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Fan, Zichen; Zhang, Qirui; An, Hyochan; Xu, Boxun; Xu, Li; Tseng, Chien-Wei; Peng, Yimai; Bejarano-Carbo, Andrea; Abillama, Pierre; Cao, Ang; Liu, Bowen; Lee, Changwoo; Wang, Zhehong; Kim, Hun-Seok; Blaauw, David; Sylvester, Dennis

被引用次数: 0

2024

A 22nm 9.51 TOPS/W Neural Engine with 2MB MRAM Leveraging Sparse-Orthogonal Walsh-Hadamard Transform Computations and Dynamic Power Gating

2024 50TH IEEE EUROPEAN SOLID-STATE ELECTRONICS RESEARCH CONFERENCE, ESSERC 2024

Abillama, Pierre; Zhang, Qirui; Fan, Zichen; An, Hyochan; Chen, Kuan-Yu; Chen, Yu; Blaauw, David; Sylvester, Dennis; Kim, Hun-Seok

被引用次数: 0

2024

mSAIL: Milligram-Scale Multi-Modal Sensor Platform for Monarch Butterfly Migration Tracking

COMMUNICATIONS OF THE ACM

Lee, Inhee; Hsiao, Roger; Carichner, Gordy; Hsu, Chin-Wei; Yang, Mingyu; Shoori, Sara; Ernst, Katherine; Carichner, Tess; Li, Yuyang; Lim, Jaechan; Julick, Cole R.; Moon, Eunseong; Sun, Yi; Phillips, Jamie; Montooth, Kristi L.; Green, Delbert A., II; Kim, Hun-Seok; Blaauw, David

被引用次数: 0

2024

RoboVisio: A Micro-Robot Vision Domain-Specific SoC for Autonomous Navigation Enabling Fully-on-Chip Intelligence via 2-MB eMRAM

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Zhang, Qirui; Fan, Zichen; An, Hyochan; Wang, Zhehong; Li, Ziyun; Wang, Guanru; Abillama, Pierre; Kim, Hun-Seok; Blaauw, David; Sylvester, Dennis

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2024

A Sub-mm³ Wireless Neural Stimulator IC for Visual Cortical Prosthesis With Optical Power Harvesting and 7.5-kb/s Data Telemetry

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lee, Jungho; Letner, Joseph G.; Lim, Jongyup; Atzeni, Gabriele; Liao, Jiawei; Kamboj, Abhilasha; Mani, Bhavika; Jeong, Seokhyeon; Kim, Yejoong; Sun, Yi; Koo, Beomseo; Richie, Julianna; Valle, Elena della; Patel, Paras R.; Sylvester, Dennis; Kim, Hun-Seok; Jang, Taekwang; Phillips, Jamie D.; Chestek, Cynthia A.; Weiland, James; Blaauw, David

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2024

A 1.5- μ W Fully-Integrated Keyword Spotting SoC in 28-nm CMOS With Skip-RNN and Fast-Settling Analog Frontend for Adaptive Frame Skipping

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Yang, Heejin; Seol, Ji-Hwan; Rothe, Rohit; Fan, Zichen; Zhang, Qirui; Kim, Hun-Seok; Blaauw, David; Sylvester, Dennis

被引用次数: 2

2024

A Low-Power Highly Reconfigurable Analog FIR Filter With 11-Bit Charge-Domain DAC for Narrowband Receivers

IEEE SOLID-STATE CIRCUITS LETTERS

Tseng, Chien-Wei; Feng, Zhen; Fan, Zichen; An, Hyochan; Wang, Yunfan; Kim, Hun-Seok; Blaauw, David

被引用次数: 0

2024

Measurement of Photoneutrons from Depleted Uranium and Comparison Study Using MCNPX-PoliMi

NUCLEAR SCIENCE AND ENGINEERING

Jinia, Abbas J.; Maurer, Tessa E.; Meert, Christopher A.; Clarke, Shaun D.; Kim, Hun-Seok; Wentzloff, David D.; Pozzi, Sara A.

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2024

An Ultralow-Power H.264/AVC Intra-Frame Image Compression Accelerator for Intelligent Event-Driven IoT Imaging Systems

IEEE SOLID-STATE CIRCUITS LETTERS

Zhang, Qirui; An, Hyochan; Bejarano-Carbo, Andrea; Kim, Hun-Seok; Blaauw, David; Sylvester, Dennis

被引用次数: 0

2024

Quantum Circuit Simulation with Fast Tensor Decision Diagram

2024 25TH INTERNATIONAL SYMPOSIUM ON QUALITY ELECTRONIC DESIGN, ISQED 2024

Zhang, Qirui; Saligane, Mehdi; Kim, Hun-Seok; Blaauw, David; Tzimpragos, Georgios; Sylvester, Dennis

被引用次数: 0

2024

A 33.06-Gb/s Reconfigurable Galois Field oFEC Decoder for Optical Intersatellite Communication

IEEE SOLID-STATE CIRCUITS LETTERS

Wei, Xiangdong; Yue, Yufan; Choi, Seungkyu; Ajayi, Tutu; Dreslinski, Ronald; Blaauw, David; Kim, Hun-Seok

被引用次数: 0

2024

ParaBase: A Configurable Parallel Baseband Processor for Ultra-High-Speed Inter-Satellite Optical Communications

PROCEEDINGS OF THE 29TH ACM/IEEE INTERNATIONAL SYMPOSIUM ON LOW POWER ELECTRONICS AND DESIGN, ISLPED 2024

Choi, Seungkyu; Deng, Huanshihong; Chen, Kuan-Yu; Yue, Yufan; Blaauw, David; Kim, Hun-Seok

被引用次数: 0

2024

Global Localization of Energy-Constrained Miniature RF Emitters using Low Earth Orbit Satellites

PROCEEDINGS OF THE 21ST ACM CONFERENCE ON EMBEDDED NETWORKED SENSOR SYSTEMS, SENSYS 2023

Wang, Yunfan; Young, Steve; Komma, Demba; Lim, Jaechan; Feng, Zhen; Fan, Zichen; Tseng, Chien-Wei; Kim, Hun-Seok; Blaauw, David

被引用次数: 1

2023

Deep Joint Source-Channel Coding with Iterative Source Error Correction

INTERNATIONAL CONFERENCE ON ARTIFICIAL INTELLIGENCE AND STATISTICS, VOL 206

Lee, Changwoo; Hu, Xiao; Kim, Hun-Seok

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2023

TaskFusion: An Efficient Transfer Learning Architecture with Dual Delta Sparsity for Multi-Task Natural Language Processing

PROCEEDINGS OF THE 2023 THE 50TH ANNUAL INTERNATIONAL SYMPOSIUM ON COMPUTER ARCHITECTURE, ISCA 2023

Fan, Zichen; Zhang, Qirui; Abillama, Pierre; Shoouri, Sara; Lee, Changwoo; Blaauw, David; Kim, Hun-Seok; Sylvester, Dennis

被引用次数: 2

2023

Deep Learning-Based Joint Channel Coding and Frequency Modulation for Low Power Connectivity

ICC 2023-IEEE INTERNATIONAL CONFERENCE ON COMMUNICATIONS

Chang, Boxuan; Wang, Chenyu; Kim, Hun-Seok

被引用次数: 0

2023

Efficient Computation Sharing for Multi-Task Visual Scene Understanding

2023 IEEE/CVF INTERNATIONAL CONFERENCE ON COMPUTER VISION (ICCV 2023)

Shoouri, Sara; Yang, Mingyu; Fan, Zichen; Kim, Hun-Seok

被引用次数: 0

2023

MMVC: Learned Multi-Mode Video Compression with Block-based Prediction Mode Selection and Density-Adaptive Entropy Coding

2023 IEEE/CVF CONFERENCE ON COMPUTER VISION AND PATTERN RECOGNITION (CVPR)

Liu, Bowen; Chen, Yu; Machineni, Rakesh Chowdary; Liu, Shiyu; Kim, Hun-Seok

被引用次数: 9

2023

MMVC: Learned Multi-Mode Video Compression with Block-based Prediction Mode Selection and Density-Adaptive Entropy Coding

IEEE/CVF Computer Vision and Pattern Recognition Conference

Bowen Liu; Yu Chen; Rakesh Chowdary Machineni; Shiyu Liu; Hun-Seok Kim

被引用次数: 0

2023

Real-Time Classification of Radiation Pulses With Piled-Up Recovery Using an FPGA-Based Artificial Neural Network

IEEE ACCESS

Michels, Noah M.; Jinia, Abbas J.; Clarke, Shaun D.; Kim, Hun-Seok; Pozzi, Sara A.; Wentzloff, David D.

被引用次数: 5

2023

Prompt Photofission Neutron Detection in Depleted Uranium

PHYSICAL REVIEW APPLIED

Jinia, A. J.; Maurer, T. E.; Meert, C. A.; Pakari, O. V.; Clarke, S. D.; Kim, H. S.; Wentzloff, D. D.; Pozzi, S. A.

被引用次数: 1

2023

Learning-Based Near-Orthogonal Superposition Code for MIMO Short Message Transmission

IEEE TRANSACTIONS ON COMMUNICATIONS

Bian, Chenghong; Hsu, Chin-Wei; Lee, Changwoo; Kim, Hun-Seok

被引用次数: 0

2023

Instantaneous Feedback-Based Opportunistic Symbol Length Adaptation for Reliable Communication

IEEE TRANSACTIONS ON COMMUNICATIONS

Hsu, Chin-Wei; Anastasopoulos, Achilles; Kim, Hun-Seok

被引用次数: 0

2023

Hyper-Dimensional Modulation for Robust Short Packets in Massive Machine-Type Communications

IEEE TRANSACTIONS ON COMMUNICATIONS

Hsu, Chin-Wei; Kim, Hun-Seok

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2023

SONA: An Accelerator for Transform-Domain Neural Networks with Sparse-Orthogonal Weights

2023 IEEE 34TH INTERNATIONAL CONFERENCE ON APPLICATION-SPECIFIC SYSTEMS, ARCHITECTURES AND PROCESSORS, ASAP

Abillama, Pierre; Fan, Zichen; Chen, Yu; An, Hyochan; Zhang, Qirui; Choi, Seungkyu; Blaauw, David; Sylvester, Dennis; Kim, Hun-Seok

被引用次数: 0

2023

Enabling Software-Defined RF Convergence with a Novel Coarse-Scale Heterogeneous Processor

2022 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS (ISCAS 22)

Bliss, D. W.; Ajayi, T.; Akoglu, A.; Aliyev, I.; Basaklar, T.; Belayneh, L.; Blaauw, D.; Brunhaver, J.; Chakrabarti, C.; Chang, L.; Chen, K-Y; Chen, M-H; Chen, X.; Chiriyath, A. R.; Daftardar, A.; Dreslinski, R.; Dutta, A.; Farcas, A. J.; Fu, Y.; Goksoy, A.; He, X.; Hassan, Md S.; Herschfelt, A.; Holtom, J.; Kim, H-S; Krishnakumar, A. N.; Li, Y.; Ma, O.; Mack, J.; Mallik, S.; Mandal, S. K.; Marculescu, R.; McCall, B.; Mudge, T.; Ogras, U. Y.; Pandey, V.; Siddiqui, S.; Sun, Y-H; Venkataramani, A.; Wei, X.; Willis, B. R.; Yu, H.; Yue, Y.

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2022

DEEP JOINT SOURCE-CHANNEL CODING FOR WIRELESS IMAGE TRANSMISSION WITH ADAPTIVE RATE CONTROL

2022 IEEE INTERNATIONAL CONFERENCE ON ACOUSTICS, SPEECH AND SIGNAL PROCESSING (ICASSP)

Yang, Mingyu; Kim, Hun-Seok

被引用次数: 59

2022

AN END-TO-END DEEP LEARNING FRAMEWORK FOR MULTIPLE AUDIO SOURCE SEPARATION AND LOCALIZATION

2022 IEEE INTERNATIONAL CONFERENCE ON ACOUSTICS, SPEECH AND SIGNAL PROCESSING (ICASSP)

Chen, Yu; Liu, Bowen; Zhang, Zijian; Kim, Hun-Seok

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2022

Efficient Deep Visual and Inertial Odometry with Adaptive Visual Modality Selection

COMPUTER VISION, ECCV 2022, PT XXXVIII

Yang, Mingyu; Chen, Yu; Kim, Hun-Seok

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2022

OFDM-Guided Deep Joint Source Channel Coding for Wireless Multipath Fading Channels

IEEE TRANSACTIONS ON COGNITIVE COMMUNICATIONS AND NETWORKING

Yang, Mingyu; Bian, Chenghong; Kim, Hun-Seok

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2022

A Power-Efficient Brain-Machine Interface System With a Sub-mw Feature Extraction and Decoding ASIC Demonstrated in Nonhuman Primates

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

An, Hyochan; Nason-Tomaszewski, Samuel R.; Lim, Jongyup; Kwon, Kyumin; Willsey, Matthew S.; Patil, Parag G.; Kim, Hun-Seok; Sylvester, Dennis; Chestek, Cynthia A.; Blaauw, David

被引用次数: 12

2022

A low-power communication scheme for wireless, 1000 channel brain-machine interfaces

JOURNAL OF NEURAL ENGINEERING

Costello, Joseph T.; Nason-Tomaszewski, Samuel R.; An, Hyochan; Lee, Jungho; Mender, Matthew J.; Temmar, Hisham; Wallace, Dylan M.; Lim, Jongyup; Willsey, Matthew S.; Patil, Parag G.; Jang, Taekwang; Phillips, Jamie D.; Kim, Hun-Seok; Blaauw, David; Chestek, Cynthia A.

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2022

A Light-Tolerant Wireless Neural Recording IC for Motor Prediction With Near-Infrared-Based Power and Data Telemetry

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lim, Jongyup; Lee, Jungho; Moon, Eunseong; Barrow, Michael; Atzeni, Gabriele; Letner, Joseph G.; Costello, Joseph T.; Nason, Samuel R.; Patel, Paras R.; Sun, Yi; Patil, Parag G.; Kim, Hun-Seok; Chestek, Cynthia A.; Phillips, Jamie; Blaauw, David; Sylvester, Dennis; Jang, Taekwang

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2022

Versa: A 36-Core Systolic Multiprocessor With Dynamically Reconfigurable Interconnect and Memory

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Sung; Fayazi, Morteza; Daftardar, Alhad; Chen, Kuan-Yu; Tan, Jielun; Pal, Subhankar; Ajayi, Tutu; Xiong, Yan; Mudge, Trevor; Chakrabarti, Chaitali; Blaauw, David; Dreslinski, Ronald; Kim, Hun-Seok

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2022

Improving Energy Efficiency of Convolutional Neural Networks on Multi-core Architectures through Run-time Reconfiguration

2022 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS (ISCAS 22)

Xiong, Y.; Li, J.; Blaauw, D.; Kim, H-S; Mudge, T.; Dreslinski, R.; Chakrabarti, C.

被引用次数: 0

2022

Squaring the circle: Executing Sparse Matrix Computations on FlexTPU-a TPU-like processor

PROCEEDINGS OF THE 2022 31ST INTERNATIONAL CONFERENCE ON PARALLEL ARCHITECTURES AND COMPILE TECHNIQUES, PACT 2022

He, Xin; Chen, Kuan-Yu; Feng, Siying; Kim, Hun-Seok; Blaauw, David; Dreslinski, Ronald; Mudge, Trevor

被引用次数: 3

2022

TRACKING THE MIGRATION OF THE MONARCH BUTTERFLIES WITH THE WORLD'S SMALLEST COMPUTER

GETMOBILE-MOBILE COMPUTING & COMMUNICATIONS REVIEW

Lee, Inhee; Hsiao, Roger; Carichner, Gordy; Hsu, Chin-Wie; Yang, Mingyu; Shoouri, Sara; Ernst, Katherine; Carichner, Tess; Li, Yuyang; Lim, Jaechan; Julick, Cole R.; Moon, Eunseong; Sun, Yi; Phillips, Jamie; Montooth, Kristi L.; Green, Delbert A.; Kim, Hun-Seok; Blaauw, David

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2022

An Ultra-Low-Power Image Signal Processor for Hierarchical Image Recognition With Deep Neural Networks

IEEE JOURNAL OF SOLID-STATE CIRCUITS

An, Hyochan; Schiferl, Sam; Venkatesan, Siddharth; Wesley, Tim; Zhang, Qirui; Wang, Jingcheng; Choo, Kyojin D.; Liu, Shiyu; Liu, Bowen; Li, Ziyun; Gong, Luyao; Zhong, Hengfei; Blaauw, David; Dreslinski, Ronald; Kim, Hun Seok; Sylvester, Dennis

被引用次数: 11

2021

Deep Learning in Latent Space for Video Prediction and Compression

2021 IEEE/CVF CONFERENCE ON COMPUTER VISION AND PATTERN RECOGNITION, CVPR 2021

Liu, Bowen; Chen, Yu; Liu, Shiyu; Kim, Hun-Seok

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2021

Instantaneous Feedback-based Opportunistic Symbol Length Adaptation for Reliable Communication

2021 IEEE GLOBAL COMMUNICATIONS CONFERENCE (GLOBECOM)

Hsu, Chin-Wei; Anastasopoulos, Achilleas; Kim, Hun-Seok

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2021

Bridging the Last Millimeter Gap of Brain-Machine Interfaces via Near-Infrared Wireless Power Transfer and Data Communications

ACS PHOTONICS

Moon, Eunseong; Barrow, Michael; Lim, Jongyup; Lee, Junggho; Nason, Samuel R.; Costello, Joseph; Kim, Hun Seok; Chestek, Cynthia; Jang, Taekwang; Blaauw, David; Phillips, Jamie D.

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2021

An Artificial Neural Network System for Photon-Based Active Interrogation Applications

IEEE ACCESS

Jinia, Abbas J.; Maurer, Tessa E.; Meert, Christopher A.; Hua, Michael Y.; Clarke, S. D.; Kim, Hun-Seok; Wentzloff, David D.; Pozzi, Sara A.

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2021

RRAM-DNN: An RRAM and Model-Compression Empowered All-Weights-On-Chip DNN Accelerator

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Li, Ziyun; Wang, Zhehong; Xu, Li; Dong, Qing; Liu, Bowen; Su, Chin-I; Chu, Wen-Ting; Tsou, George; Chih, Yu-Der; Chang, Tsung-Yung Jonathan; Sylvester, Dennis; Kim, Hun-Seok; Blaauw, David

被引用次数: 24

2021

Physical Layer Secret Key Generation Using Joint Interference and Phase Shift Keying Modulation

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Ebrahimi, Najme; Kim, Hun-Seok; Blaauw, David

被引用次数: 12

2021

Millimeter-Scale Node-to-Node Radio Using a Carrier Frequency-Interlocking IF Receiver for a Fully Integrated 4 $\times$ 4 $\times$ 4 mm³ Wireless Sensor Node

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Chuo, Li-Xuan; Feng, Zhen; Kim, Yejoong; Chiotellis, Nikolaos; Yasuda, Makoto; Miyoshi, Satoru; Kawaminami, Masaru; Grbic, Anthony; Wentzloff, David; Blaauw, David; Kim, Hun-Seok

被引用次数: 14

2020

Minimax particle filtering for tracking a highly maneuvering target

INTERNATIONAL JOURNAL OF ROBUST AND NONLINEAR CONTROL

Lim, Jaechan; Kim, Hun-Seok; Park, Hyung-Min

被引用次数: 12

2020

A low-power band of neuronal spiking activity dominated by local single units improves the performance of brain-machine interfaces

NATURE BIOMEDICAL ENGINEERING

Nason, Samuel R.; Vaskov, Alex K.; Willsey, Matthew S.; Welle, Elissa J.; An, Hyochan; Vu, Philip P.; Bullard, Autumn J.; Nu, Chrono S.; Kao, Jonathan C.; Shenoy, Krishna, V; Jang, Taekwang; Kim, Hun-Seok; Blaauw, David; Patil, Parag G.; Chestek, Cynthia A.

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2020

A 7.3 M Output Non-Zeros/J, 11.7 M Output Non-Zeros/GB Reconfigurable Sparse Matrix-Matrix Multiplication Accelerator

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Park, Dong-Hyeon; Pal, Subhankar; Peng, Siying; Gao, Paul; Tan, Jielun; Rovinski, Austin; Xie, Shaolin; Zhao, Chun; Amarnath, Aporva; Wesley, Timothy; Beaumont, Jonathan; Chen, Kuan-Yu; Chakrabarti, Chaitali; Taylor, Michael Bedibrd; Mudge, Trevor; Blaauw, David; Kim, Hun-Seok; Dreslinski, Ronald G.

被引用次数: 19

2020

Interactive-Multiple-Model Algorithm Based on Minimax Particle Filtering

IEEE SIGNAL PROCESSING LETTERS

Lim, Jaechan; Kim, Hun-Seok; Park, Hyung-Min

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2020

An Acoustic Signal Processing Chip With 142-nW Voice Activity Detection Using Mixer-Based Sequential Frequency Scanning and Neural Network Classification

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Oh, Sechang; Cho, Minchang; Shi, Zhan; Lim, Jongyup; Kim, Yejoong; Jeong, Seokhyeon; Chen, Yu; Rothe, Rohit; Blaauw, David; Kim, Hun-Seok; Sylvester, Dennis

被引用次数: 41

2019

Enhanced Interference Rejection Bluetooth Low-Energy Back-Channel Receiver With LO Frequency Hopping

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Alghaihab, Abdullah; Shi, Yao; Breiholz, Jacob; Kim, Hun-Seok; Calhoun, Benton H.; Wentzloff, David D.

被引用次数: 18

2019

Analysis and Design of an Ultra-Low-Power Bluetooth Low-Energy Transmitter With Ring Oscillator-Based ADPLL and 4x Frequency Edge Combiner

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Chen, Xing; Breiholz, Jacob; Yahya, Farah B.; Lukas, Christopher J.; Kim, Hun-Seok; Calhoun, Benton H.; Wentzloff, David D.

被引用次数: 56

2019

A 1920 x 1080 25-Frames/s 2.4-TOPS/W Low-Power 6-D Vision Processor for Unified Optical Flow and Stereo Depth With Semi-Global Matching

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Li, Ziyun; Wang, Jingcheng; Sylvester, Dennis; Blaauw, David; Kim, Hun Seok

被引用次数: 23

2019

Low Complexity, Hardware-Efficient Neighbor-Guided SGM Optical Flow for Low-Power Mobile Vision Applications

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY

Li, Ziyun; Xiang, Jiang; Gong, Luyao; Blaauw, David; Chakrabarti, Chaitali; Kim, Hun Seok

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2019

A 220- μ W-83-dBm 5.8-GHz Third-Harmonic Passive Mixer-First LP-WUR for IEEE 802.11ba

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Im, Jaeho; Kim, Hun-Seok; Wentzloff, David D.

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2019

Always-On 12-nW Acoustic Sensing and Object Recognition Microsystem for Unattended Ground Sensor Nodes

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jeong, Seokhyeon; Chen, Yu; Jang, Taekwang; Tsai, Julius Ming-Lin; Blaauw, David; Kim, Hun-Seok; Sylvester, Dennis

被引用次数: 59

2018

Artificial neural network algorithms for pulse shape discrimination and recovery of piled-up pulses in organic scintillators

ANNALS OF NUCLEAR ENERGY

Fu, C.; Di Fulvio, A.; Clarke, S. D.; Wentzloff, D.; Pozzi, S. A.; Kim, H. S.

被引用次数: 58

2018

A Fixed-Point Neural Network Architecture for Speech Applications on Resource Constrained Hardware

JOURNAL OF SIGNAL PROCESSING SYSTEMS FOR SIGNAL IMAGE AND VIDEO TECHNOLOGY

Shah, Mohit; Arunachalam, Sairam; Wang, Jingcheng; Blaauw, David; Sylvester, Dennis; Kim, Hun-Seok; Seo, Jae-sun; Chakrabarti, Chaitali

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2018

A 1920 x 1080 30-frames/s 2.3 TOPS/W Stereo-Depth Processor for Energy-Efficient Autonomous Navigation of Micro Aerial Vehicles

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Li, Ziyun; Dong, Qing; Saligane, Mehdi; Kempke, Benjamin; Gong, Luyao; Zhang, Zhengya; Dreslinski, Ronald; Sylvester, Dennis; Blaauw, David; Kim, Hun-Seok

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2018

Analysis of Circuit Noise and Non-Ideal Filtering Impact on Energy Detection Based Ultra-Low-Power Radios Performance

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Alghaihab, Abdullah Mohammed; Kim, Hun-Seok; Wentzloff, David D.

被引用次数: 10

2018

A MURS Band Digital Quadrature Transmitter With Class-B I/Q Cell Sharing for Long Range IoT Applications

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Kosari, Avish; Kim, Hun-Seok; Wentzloff, David D.

被引用次数: 4

2018

Circuit and System Designs of Ultra-Low Power Sensor Nodes With Illustration in a Miniaturized GNSS Logger for Position Tracking: Part I-Analog Circuit Techniques

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Jang, Taekwang; Kim, Gyouho; Kempke, Benjamin; Henry, Michael B.; Chiotellis, Nikolaos; Pfeiffer, Carl; Kim, Dongkwun; Kim, Yejoong; Foo, Zhiyoong; Kim, Hyeongseok; Grbic, Anthony; Sylvester, Dennis; Kim, Hun-Seok; Wentzloff, David D.; Blaauw, David

被引用次数: 27

2017

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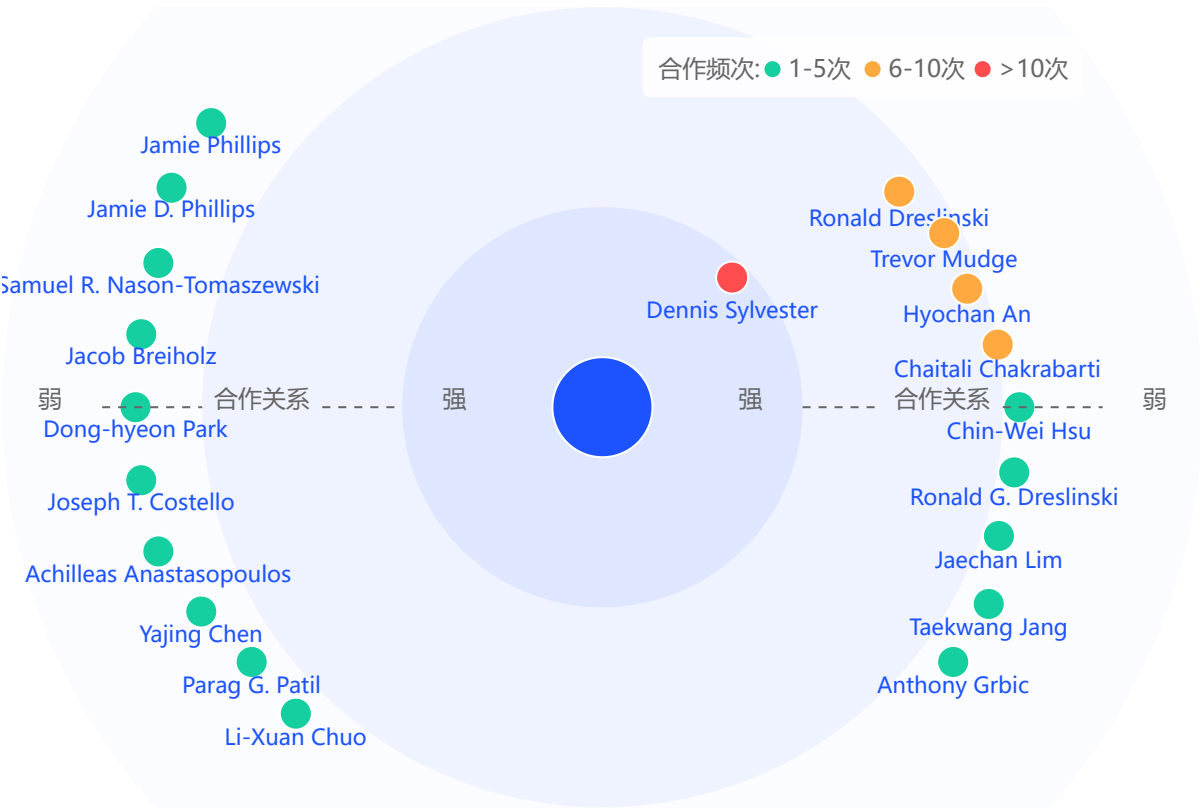
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